# Spatiotemporal Graph Neural Networks for Neighborhood-Level Influenza Forecasting in Boston


## Abstract

Health departments act neighborhood by neighborhood, but most influenza forecasts cover states or regions. We asked whether a map of how neighborhoods connect improves neighborhood forecasts. We forecast emergency department influenza-like illness 1, 2, and 4 weeks ahead in Boston, Massachusetts (14 neighborhoods, 2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (58 ZIP codes, 2022–2026). Seasons used to design the model were excluded, and forecasts were scored with the weighted interval score used by FluSight. In Boston, a graph neural network had 18% to 28% less error than assuming next week equals this week, and removing the map increased its error in 8 of 9 season and horizon comparisons (clearly in 5). A tuned recurrent neural network without a map matched it 1 week ahead; the graph neural network's lead at 2 and 4 weeks was clear in 1 of 3 seasons. The map could be built from public boundary files, and a Boston-trained model forecast Columbus as well as a local model. In Chicago, never used in design, error was 17% to 19% below persistence, and the map helped 1 week ahead in all 4 seasons. The map's gain was modest but measurable.



**Keywords:** forecasting; graph neural networks; influenza, human; machine learning; public health surveillance; small-area analysis


**Abbreviations:** ED, emergency department; ILI, influenza-like illness; LSTM, long short-term memory network; WIS, weighted interval score.

# Introduction

Seasonal influenza affects up to 1 billion people worldwide each year,[1] and in the United States its annual economic burden has been estimated at $11.2 billion.[2] That burden is not shared evenly within a city. During the 1918 pandemic, influenza mortality in Chicago differed from one census tract to the next and tracked illiteracy and homeownership.[3] A century later, influenza hospitalization rates in US census tracts where at least 20% of residents lived in poverty were nearly twice those in tracts where fewer than 5% did.[4] In our data, from June 2022 to May 2025, the average weekly rate of emergency department (ED) visits for influenza-like illness (ILI) ranged from 9 per 100,000 residents in Boston's Fenway neighborhood to 67 per 100,000 in Dorchester.

Health departments act on that variation locally. Mobile vaccination clinics, school outreach, and messaging to clinicians are placed neighborhood by neighborhood. Forecasts help when a decision has to be made before the data that would settle it arrive,[5] and public health agencies, including the Centers for Disease Control and Prevention, have used influenza forecasts for risk communication and planning.[6] Most forecasting effort, however, targets nations, regions, and states. The agency's FluSight challenge has compared forecasts of state and national influenza activity for over a decade,[7–9] and internet search methods estimate current activity rather than forecast it[10–12] and can drift.[13]

Forecasting within a city is harder because each area has fewer cases and noisier weekly counts. It is also where knowledge of neighboring areas might help most: influenza spreads between nearby places[14] and along commuting and contact networks.[15–17] Yang et al. forecast influenza in New York City boroughs and neighborhoods with a network model and found that linking areas improved borough forecasts but not neighborhood forecasts.[18] Graph neural networks, machine learning models that pass information along a map of connections, offer another way to use this structure.[19,20] Luo et al. reported that a graph neural network with both geographic and correlation-based connections improved influenza estimates across regions of Taiwan.[21]

The evidence that such models help in practice is mixed. In multimodel forecasting hubs for influenza and COVID-19, individual models' skill varied widely, and in some seasons only a minority beat a simple baseline,[9,22] while plain autoregressive models remain competitive 1 to 4 weeks ahead.[23] In a study of dengue in Sri Lanka, a per-district recurrent neural network outperformed every graph neural network variant that weighted neighboring districts.[24] Reviews note that the added value of artificial intelligence for epidemics is hard to evaluate[25] and that retrospective tests can overstate real-world accuracy.[26]

We therefore asked a predictive question, not one about the causes of influenza: when does a neighborhood map improve forecasts of neighborhood ED visit rates for ILI in Boston 1 to 4 weeks ahead? The main comparison removes the map and holds everything else fixed, within an evaluation built to avoid the usual ways a new model can look better than it is: unseen test seasons, equal tuning for every model, and week-level tests on full forecast distributions. We also asked whether the map can be built automatically from public boundary files, and whether a model trained in one city can forecast another. We repeated the analysis in Columbus, Ohio, and in Chicago, Illinois, a city never used while designing the model, whose health department reports a 1–10 activity level for each ZIP code.

# Methods

## Setting and data

Table 1 summarizes the 3 cities. In Boston, the Boston Public Health Commission publishes the weekly rate of ED visits for ILI per 100,000 residents for 14 neighborhoods;[27] we used December 2017 through May 2026. ED visits track laboratory-confirmed influenza closely, including by age group.[28] Weeks with small counts are suppressed by the commission; we treated them as missing rather than as zero. In Columbus, Columbus Public Health provided visit-level ED records for ILI beginning in January 2022, which we aggregated to the department's 17 planning areas through a ZIP code crosswalk and divided by American

Community Survey population.[29] In Chicago, the Department of Public Health publishes, from September 2019, a weekly ILI activity level for each ZIP code from 1 to 10, comparing its share of ED visits for ILI with its own summer average;[30] Chicago errors are therefore in levels (Table 1; Appendix S1).

Each area also received 6 weekly weather variables from the Open-Meteo archive,[31] because humidity and temperature shape influenza transmission,[32–34] and 8 fixed demographic measures (Appendix S1). These were included because age and socioeconomic position shape who becomes ill.[4,35]

Boston and Chicago data are public weekly aggregates. Columbus provided deidentified visit-level records, which we aggregated to weekly area counts before analysis. [Reviewing body (blinded), determination and number; any data use agreement.]

## The neighborhood map

We represented each city as a map in which every area is a point and 2 areas are linked when they share a border (Figure 1A). Areas at the edge of a city would otherwise look as if they had no neighbors on one side, so we added surrounding areas that are part of the map but are never forecast or scored: 7 in Boston (for example, Cambridge and Somerville, Brookline, and Boston Harbor), 5 in Columbus, and 1 in Chicago (Appendix S1).

Boston's map was first drawn by hand from a city map. To test whether a new city could skip that step, we also built it automatically from public boundary files,[36–38] treating 2 areas as neighbors when their boundaries share an edge rather than only a corner. Building neighbor lists from census boundary files is standard in spatial disease mapping, as is adding links across water by hand.[39] Links across the Charles River and Boston Harbor were added by hand in every version. In addition to shared borders, which also reach 2 and 3 borders away at decreasing weight, the model's default map links neighborhoods whose past influenza curves were highly correlated (computed on weeks before the test season only) and neighborhoods with similar demographics.

## Forecasting model

The graph neural network forecasts each neighborhood in 3 steps. First, it reads the past 16 weeks of that neighborhood's ILI rate, together with weather and demographics, and summarizes, from past weeks only, whether activity is rising or falling. Second, it mixes each neighborhood's summary with those of its neighbors on the map, a step known as graph convolution.[19] The model can also learn additional links that the map does not contain. Third, it adds a learned correction to a baseline that blends 2 simple forecasts: the current week's rate and the usual rate for that time of year, estimated from training seasons with a smooth seasonal curve. The weight on each part of the blend is learned separately for each forecast horizon (how many weeks ahead), so that 1-week forecasts can lean on the current week and 4-week forecasts on the seasonal pattern (Appendix S1, Figure S1). A separate model was trained for each horizon, and each forecast averages 10 models trained from different random starting points.

Each model produced a full forecast distribution, reported as the 23 quantiles (percentiles) FluSight uses.[9] All models except the FluSight baseline, whose ranges come from past changes, got their ranges from one shared procedure calibrated on validation weeks (the weeks just before each test season).

To measure what the map contributes, we trained an otherwise identical model with the map removed (the no-map version): no links between neighborhoods, mapped or learned, and no surrounding areas, so each neighborhood sees only its own history.

## Comparison models

Appendix S1 details all comparison models, including repeating the same week of the previous year. Persistence assumes next week will look like this week. The FluSight baseline carries the latest value forward with ranges based on past changes over the forecast horizon.[9,22] The seasonal blend is the graph neural network's own baseline with the learned correction switched off, so it involves no machine learning. We also fit a seasonal autoregres-

sive integrated moving average model for each neighborhood; gradient-boosted regression trees (many small decision trees, each correcting the errors of the last) on the same inputs as the graph neural network; a long short-term memory network (LSTM), a recurrent neural network that reads each series week by week and carries a memory forward, here reading all neighborhoods' series together but with no map;[40] and Cola-GNN, a published graph neural network for influenza that learns its own connections between locations,[41] which we reimplemented from its description. Two graph models from Luo et al.[21,42] were run at 1 week only and are reported in Appendix S1.

## Evaluation design

We used rolling-origin evaluation, which reproduces how a forecaster would have performed in past seasons.[43] For each test season, every model was trained only on data before that season, tuned on the weeks immediately before it, and then scored on its 52 target weeks (53 in Chicago 2025–2026).

The 2025–2026 season in Boston and Columbus was used while designing the model, so we call it the development season and do not report it as evidence; its ablations are in Appendix S1, Table S16. The remaining seasons were never used to choose anything: 2022–2023, 2023–2024, and 2024–2025 in Boston and 2024–2025 in Columbus. Chicago was added after the design was fixed, so all 4 of its seasons, 2022–2023 through 2025–2026, are unseen (Figure 1B). Influenza nearly disappeared during the first year of the COVID-19 pandemic,[44] so Columbus models used weeks from July 2022 onward, leaving 1 unseen season; Boston and Chicago models used their full histories.

A new model can also look better because more effort went into tuning its settings. To prevent this, every machine learning model received the same budget in every test season. We tried 12 combinations of settings (its default plus 11 drawn at random from ranges of similar breadth for every model), trained each once, kept the one with the lowest error on the validation weeks, and retrained it 10 times (Appendix S1, Table S2). The same procedure

was used in all 3 cities (Appendix S1, Table S14).

## Scoring

Our main measure answers the question: How far off was the forecast, counting both its best guess and how honest its range was? This is the weighted interval score (WIS), the scoring rule used by FluSight and the COVID-19 Forecast Hubs.[9,45] Each forecast gives a median $m$ and $K = 11$ central intervals $[l_k, u_k]$ (98% to 10%); for observed value $y$,

$$\mathrm{WIS} = \frac{1}{K + \frac{1}{2}} \left( \tfrac{1}{2}|y - m| + \sum_{k=1}^{K} \left[ \tfrac{\alpha_k}{2}(u_k - l_k) + (l_k - y)^+ + (y - u_k)^+ \right] \right),$$

where $\alpha_k$ is 1 minus the interval's coverage and $(x)^+ = \max(x, 0)$: interval widths plus penalties for observations outside them (Appendix S1 explains the score and how to read a value, with a worked example). It is expressed in the units of the forecast (ED visits per 100,000, or levels in Chicago), lower is better, and it rewards ranges that are narrow but still contain what happened. Point measures answer narrower questions: mean absolute error (how far off the best guess is, on average), root mean squared error (the same, weighting large misses more), and the correlation between forecast and observed rates in each neighborhood (whether the forecast rises and falls at the right times). None scores the range, so they are secondary. Relative WIS, a model's mean WIS divided by that of persistence, is below 1 when the model beats persistence. We checked calibration by asking how often the observed value fell inside the 50% and 95% forecast intervals.[46]

To compare 2 models within a season, we asked whether the gap between them could be explained by which weeks the season happened to contain. We averaged each model's WIS over neighborhoods for each week, estimated a 95% confidence interval for the difference by resampling blocks of 4 consecutive weeks (2,000 resamples),[47] and tested it with the Diebold–Mariano test,[48] using a small-sample correction[49] and a variance estimate that allows for the overlap between consecutive multiweek forecasts.[50] Because each model was compared

against many others, 2-sided $P$ values were adjusted with the Holm method[51] within each season and horizon. We call a difference *clear* when the adjusted $P$ value was below .05, and not clear (a tie) otherwise.

## Severity alerts

We also asked whether a forecast would have warned that a neighborhood was entering a moderate or high week. As in the Centers for Disease Control and Prevention's severity classification,[52] we set thresholds for each neighborhood with the Moving Epidemic Method,[53] using only seasons before each test season and leaving out those disrupted by COVID-19; moderate or higher means at or above the 50th-percentile intensity threshold, and high or higher at or above the 90th. We scored 2 ways of alerting: from the forecast's probability of crossing the threshold, with the Brier skill score[54] (above 0 is better than always forecasting the average share of such weeks), and from whether the best guess crossed it, as hit rate minus false-alarm rate (the Peirce skill score). Differences got block-bootstrap intervals as above. Chicago's measure is already a graded level, so it was not included.

## Portability

We ran 2 further tests with model settings fixed in advance. First, we reran the Boston evaluation with 2 automatically built maps for the graph neural network and for a borders-only version that learns no extra links, so that the map is its only spatial information. Second, a version with no neighborhood-specific learned links, which can run in any city, was trained on one city's weeks before 2024–2025 and applied to another city's 2024–2025 season as is and after brief further training on local data (fine-tuning), and compared with the same model trained from scratch there: from Boston to Columbus and Chicago, and from each to Boston. We also trained it on 2 or 3 cities at once (Appendix S1).

Analyses used Python 3.12, PyTorch 2.14, PyTorch Geometric 2.8, statsmodels 0.14, scikit-learn 1.9, and XGBoost 3.4; code and frozen settings are released with the paper.[55]

# Results

## Boston

The graph neural network had the lowest mean WIS at every horizon (Table 2): 5.13 per 100,000 at 1 week, 6.47 at 2 weeks, and 8.22 at 4 weeks, against 6.24, 7.94, and 11.44 for persistence. Its error was 18% to 19% lower than that of persistence at 1 and 2 weeks and 28% lower at 4 weeks (Figure 2A). Its advantage over persistence was clear in all 3 seasons at 1 week and in 2023–2024 at 2 and 4 weeks. The graph neural network also had the lowest mean absolute error and root mean squared error at every horizon. By correlation, the tuned LSTM and Cola-GNN were slightly higher at 1 week (Appendix S1, Table S9).

## What the neighborhood map adds

Removing the map increased the model's WIS in 8 of 9 comparisons across seasons and horizons (Figure 3A). The 95% interval for the difference excluded zero in 7 of 9 comparisons, and the difference was clear in all 3 seasons at 1 week and in 2023–2024 at 2 and 4 weeks. Averaged over the 3 seasons, the map lowered WIS in every one of the 14 neighborhoods at every horizon, by 6% to 15% at 1 week and by 2% to 8% at 2 and 4 weeks (Figure 4).

## Comparison with tuned machine learning models

Tuning helped the comparison models more than the graph neural network (across the 3 cities, validation error fell 7% for the LSTM and 2% for the graph neural network; Appendix S1, Figure S2). The tuned LSTM, which reads all 14 neighborhood series at once without a map, matched the graph neural network at 1 week in all 3 seasons (mean WIS 5.18 against 5.13; Figure 3B). At 2 and 4 weeks the graph neural network had lower average error (6.47 vs 6.82 and 8.22 vs 9.16), but the difference was clear only in 2023–2024. Against tuned Cola-GNN it had lower average error at every horizon, but no difference was clear after adjustment (Figure 3C).

## Forecast ranges and an example

Forecast ranges were well calibrated (Figure 5A; Appendix S1, Figure S3): across horizons, 93% to 94% of observed values fell inside the graph neural network's 95% intervals and 52% to 55% inside its 50% intervals, similar to the comparison models. In Dorchester, the neighborhood with the highest rates, the median forecast rose with the 2024–2025 epidemic earlier than persistence, which by construction trails the data by $h$ weeks, but fell short of the peak (Figure 5B–D; all neighborhoods in Appendix S1, Figure S4).

## Severity alerts

In Boston, 232 of 2,103 neighborhood-weeks were moderate or higher and 82 were high or higher. Read as probabilities, the graph neural network's forecasts warned of moderate weeks better than persistence at 1 and 2 weeks (Brier skill score 0.61 vs 0.51 and 0.41 vs 0.31; intervals excluded zero) and better than the no-map version at 1 week (0.61 vs 0.54), similar to the LSTM (Appendix S1, Figure S5 and Table S12). Read as yes/no alerts from the best guess, they were no better than persistence for moderate weeks and caught fewer high weeks at 1 week (Peirce skill score 0.40 vs 0.59), because they fell short of the highest peaks (Figure 6). Their first crossing of a neighborhood's moderate threshold came a median of 1 week late at 1 week ahead and 2 weeks late at 2 weeks ahead, the same as persistence (Appendix S1, Table S13). No model warned of high weeks 4 weeks ahead.

## Building the map automatically and moving between cities

Against the 2 automatically built maps, the hand-drawn Boston map had 18 or 19 of its 23 borders right, 4 or 5 that do not exist, and about 5 missed. Replacing it made little difference (Table 3): none of 36 week-level comparisons (2 maps, 3 seasons, 3 horizons, and 2 models) was clear after adjustment, including for the borders-only model.

Applied to Columbus with no local training, a Boston-trained model had a mean WIS

of 3.90, 5.08, and 6.45 at 1, 2, and 4 weeks, against 4.11, 5.18, and 6.49 for the same model trained in Columbus (Table 4), with no clear difference. In Chicago, which had nearly 5 years of its own history, it was clearly worse than a Chicago-trained model at 1 and 2 weeks and tied at 4. Models moved to Boston from Columbus or Chicago were worse than a Boston-trained model at every horizon, clearly at 4 weeks (10.07 and 10.36 vs 8.25). Every transferred model had lower mean WIS than persistence, and fine-tuning never clearly differed from training from scratch (12 comparisons). In Columbus, a model trained on Boston and Chicago, without Columbus data, beat the Boston-only model at every horizon (clearly at 2 weeks: 4.80 vs 5.08); no multicity model clearly beat a local one (Appendix S1, Table S15).

## Columbus and Chicago

In Columbus, the graph neural network had the lowest mean WIS of the models in Figure 2B at every horizon (relative WIS 0.87, 0.85, and 0.75; Appendix S1, Table S4). In this single season, its leads over persistence and the no-map version were not clear at any horizon (Appendix S1, Table S7), and at 4 weeks, repeating the same week of the previous year had lower WIS (5.75 vs 6.31). It warned of moderate weeks better than persistence on both alert scores at 1 and 2 weeks, but its yes/no alerts caught fewer high weeks, and its first crossings of the moderate threshold came later (Appendix S1, Tables S12 and S13).

In Chicago, the graph neural network had the lowest mean WIS at every horizon (relative WIS 0.81, 0.83, and 0.83; Figure 2C; Appendix S1, Table S5), clearly beat persistence in 11 of 12 season-horizon comparisons, and was not clearly worse than any model in any. Removing the map raised its error in all 12, clearly at 1 week in all 4 seasons and at 2 weeks in 2, but not at 4 weeks (Figure 3A). Unlike in Boston, it clearly beat the tuned LSTM and Cola-GNN at 1 week in 3 of 4 seasons (Figure 3; Appendix S1, Table S8).

# Discussion

In Boston, a map of how neighborhoods connect made forecasts more accurate across 3 seasons, with well-calibrated ranges and lower error in every neighborhood. The gain over strong alternatives was modest: a tuned LSTM that sees every neighborhood's series but no map matched it 1 week ahead, and the graph neural network's lead at 2 and 4 weeks (ahead in 5 of 6 comparisons) was clear in 1 season of 3. In Columbus, no gain was clear.

Linking areas also gave little at fine scales in New York City and for dengue in Sri Lanka.[18,24] In forecasting hubs, gains over simple baselines have varied by season and grown at longer horizons.[9,22] One explanation for the small margin over the LSTM is that Boston's neighborhoods rise and fall largely together. A model that reads all 14 series at once can learn much of what a neighbor's data would tell it, with or without a map. If so, the map should matter most to a model that cannot otherwise see other neighborhoods' series.

Chicago was the strongest test of portability. Designed entirely on Boston, the model carried over to a city with 4 times as many areas and a different measure, beating persistence by 17% to 19%; its clearer lead over the LSTM there fits that explanation.

For health departments, the method needs weekly area-level rates, public boundary files (plus any links across water), and a season definition. A department with a short surveillance history can start with a model trained elsewhere, as graph models have transferred between countries for COVID-19.[56] In Columbus, models from other cities matched a local one, and training on 2 cities did better than on 1; Chicago and Boston did better with their own. For alerts, a department should act on a forecast's probability of crossing a threshold rather than on its best guess, which falls short of peaks; no model warned earlier than simply carrying the latest week forward. Before relying on any forecast, a department should check it against persistence on its own past seasons. An open-source Python package runs this check and the forecasts on a department's own data; it reproduced the Boston evaluation within 0.4% mean WIS in all 27 model, season, and horizon comparisons.[57]

This study has limitations. Every model, like persistence, ran about $h$ weeks behind

the observed curve at $h$ weeks ahead; adding a growth-rate feature or training on median loss did not change this, so the gains over persistence come from forecasting the size of the epidemic, not from anticipating its turns. Models trained on past rates largely learn persistence; earlier warning needs signals that lead ED visits for epidemiologic reasons, such as wastewater, available within 24 hours of sampling,[58] rather than a larger model. Columbus contributed 1 unseen season, and Chicago's measure is a capped index rather than a rate, which flattens peaks. All models used final data, whereas real-time forecasters face reporting delays and revisions that can make retrospective accuracy look better.[59] Because all inputs shift together, our 4-week forecasts stand in for 2-week forecasts made with a 2-week reporting delay; we did not examine Boston's data revisions. The model's structure was fixed on the development season, with only training settings retuned for each test season; Cola-GNN was reimplemented from its published description; and Boston's links across water were added by hand. Finally, because WIS is measured on the scale of the rates, neighborhoods and weeks with high rates weigh most,[60] and ED visits for ILI capture people who seek emergency care, not all infections.

Next steps are a real-time test in the 2026–2027 season, now under way with weekly forecasts posted publicly (see Data Availability), ensembles of models,[61,62] and wastewater as an input.

Neighborhood-level influenza forecasts 1 to 4 weeks ahead are feasible from routine ED surveillance. In a city with a long, complete surveillance history, a neighborhood map lowers forecast error modestly in every neighborhood, and its borders can come from public boundary files.

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# Tables

Table 1: Surveillance Data and Evaluation Seasons for Neighborhood-Level Influenza Forecasting in Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, 2017–2026

| Characteristic | Boston | Columbus | Chicago |
|---|---|---|---|
| Areas forecast | 14 neighborhoods | 17 planning areas | 58 ZIP codes |
| Surrounding areas, not scored | 7 | 5 | 1 |
| Measure | ED visits for ILI per 100,000 | ED visits for ILI per 100,000 | ILI activity level, 1–10[a] |
| Source | Boston Public Health Commission | Columbus Public Health, visit-level records | Chicago Department of Public Health |
| Population denominator | Published rate | American Community Survey | None (index) |
| Data begin | December 2017 | January 2022[b] | September 2019 |
| Last week used | May 2026 | May 2026 | May 2026 |
| Peak months | October–March | October–March | October–March |
| Test year | June–May | June–May | June–May |
| Unseen test seasons | 2022–2023, 2023–2024, 2024–2025 | 2024–2025 | 2022–2023 to 2025–2026 |
| Development season[c] | 2025–2026 | 2025–2026 | None |
| Neighborhood map | Hand drawn; also built from boundary files | Built from boundary files | Built from boundary files |
| Demographic measures | 8 | 8 | 8 |

Abbreviations: ED, emergency department; ILI, influenza-like illness.

[a] A ZIP code's share of ED visits for ILI against its own summer average, in steps of about 1 standard deviation and capped at 10.

[b] Columbus models used weeks from July 2022 onward.

[c] Used to design the model and not reported as evidence.

Table 2: Forecast Error by Model and Horizon for 14 Neighborhoods, Boston, Massachusetts, 3 Unseen Seasons, June 2022–May 2025

| | Mean WIS[a] | | | Relative WIS[b] | | | Mean Absolute Error[a] | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | 1 wk | 2 wk | 4 wk | 1 wk | 2 wk | 4 wk | 1 wk | 2 wk | 4 wk |
| Graph neural network | 5.13 | 6.47 | 8.22 | 0.82 | 0.81 | 0.72 | 7.85 | 9.63 | 11.93 |
| Same model, map removed | 5.69 | 6.72 | 8.68 | 0.91 | 0.85 | 0.76 | 8.76 | 10.10 | 12.57 |
| LSTM | 5.18 | 6.82 | 9.16 | 0.83 | 0.86 | 0.80 | 7.96 | 10.19 | 13.02 |
| Cola-GNN | 5.35 | 7.61 | 8.77 | 0.86 | 0.96 | 0.77 | 8.19 | 11.04 | 12.27 |
| Gradient-boosted trees | 6.08 | 7.32 | 9.05 | 0.97 | 0.92 | 0.79 | 9.21 | 10.82 | 12.99 |
| Seasonal ARIMA | 6.52 | 7.66 | 9.15 | 1.04 | 0.97 | 0.80 | 9.69 | 11.17 | 13.09 |
| Seasonal blend, no learning | 5.95 | 7.29 | 9.57 | 0.95 | 0.92 | 0.84 | 9.06 | 10.88 | 13.69 |
| FluSight baseline | 6.39 | 8.16 | 11.45 | 1.02 | 1.03 | 1.00 | 9.50 | 11.89 | 16.23 |
| Persistence | 6.24 | 7.94 | 11.44 | 1.00 | 1.00 | 1.00 | 9.48 | 11.89 | 16.22 |

Abbreviations: ARIMA, autoregressive integrated moving average; ED, emergency department; LSTM, long short-term memory network; WIS, weighted interval score; wk, weeks ahead.
[a] ED visits for influenza-like illness per 100,000 residents; mean of the 3 season means; lower is better. Learned models were tuned with an equal budget and averaged over 10 training runs.
[b] Mean WIS divided by that of persistence; values below 1 indicate less error than persistence.

Table 3: Forecast Error With Hand-Drawn vs Automatically Built Neighborhood Maps, Boston, Massachusetts, 3 Unseen Seasons, June 2022–May 2025

| | | Mean WIS[a] by Map | | |
|---|---|---|---|---|
| Model | Weeks Ahead | Hand Drawn | Land Boundaries[b] | Census Tracts[c] |
| Graph neural network | 1 | 5.02 | 5.04 | 5.03 |
| | 2 | 6.28 | 6.24 | 6.39 |
| | 4 | 8.31 | 8.26 | 8.26 |
| Borders only[d] | 1 | 5.10 | 5.13 | 5.07 |
| | 2 | 6.38 | 6.39 | 6.33 |
| | 4 | 8.40 | 8.35 | 8.29 |

Abbreviations: ED, emergency department; WIS, weighted interval score.
[a] ED visits for influenza-like illness per 100,000 residents. Models use settings fixed before evaluation, without per-season tuning, so values differ slightly from Table 2.
[b] Official neighborhood land boundaries, matched to surrounding towns within 100 m.
[c] Neighborhoods approximated by 2020 census tracts.
[d] Uses shared borders only and learns no additional links, so the map is its only spatial information.

Table 4: Forecast Error When a Model Trained in One City Is Applied to Another, Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, Unseen Season June 2024–May 2025

| Target City (Source) | Weeks Ahead | Mean WIS[a] | | | |
|---|---|---|---|---|---|
| | | Trained Locally[b] | Transferred, No Local Training | Transferred and Fine-Tuned | Persistence |
| Columbus (from Boston) | 1 | 4.11 | 3.90 | 4.10 | 4.67 |
| | 2 | 5.18 | 5.08 | 5.09 | 5.89 |
| | 4 | 6.49 | 6.45 | 6.49 | 8.45 |
| Chicago (from Boston)[c] | 1 | 1.09 | 1.16 | 1.08 | 1.35 |
| | 2 | 1.12 | 1.24 | 1.12 | 1.42 |
| | 4 | 1.25 | 1.29 | 1.23 | 1.52 |
| Boston (from Columbus) | 1 | 4.88 | 5.28 | 4.78 | 6.36 |
| | 2 | 6.15 | 7.11 | 6.27 | 8.95 |
| | 4 | 8.25 | 10.07 | 8.37 | 14.07 |
| Boston (from Chicago) | 1 | 4.88 | 5.15 | 4.87 | 6.36 |
| | 2 | 6.15 | 6.86 | 6.12 | 8.95 |
| | 4 | 8.25 | 10.36 | 8.41 | 14.07 |

Abbreviations: ED, emergency department; WIS, weighted interval score.

[a] In the target city's units: ED visits for influenza-like illness per 100,000 residents in Boston and Columbus, and the health department's 1–10 activity level in Chicago. Compare columns within a row, not values across cities. The model has no neighborhood-specific learned links, so its parameters have the same size in every city, and uses settings fixed before evaluation. The source model was trained only on weeks before the target's test season.

[b] The same no-learned-links model trained on the target city alone. It is not the tuned graph neural network of Table 2, so its values differ from that table.

[c] Chicago's measure is a level rather than a rate. Each area's series is scaled by its own training-period mean and standard deviation, so the transferred model forecasts in standardized units that are converted back to levels.

# Figure Legends

**Figure 1.** Study design for neighborhood-level influenza forecasting in Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, 2017–2026. A) Boston's 14 neighborhoods, shaded by mean weekly rate of emergency department visits for influenza-like illness per 100,000 residents, June 2022–May 2025. Solid lines link neighborhoods that share a border in the official land boundary file; dashed lines are links across the Charles River and Boston Harbor, which were added by hand. Other surrounding areas are not shown. B) Data used in each city and the test seasons. Each model was trained only on data before a test season and scored on its target weeks. The development season, used to design the model, is not reported as evidence; Chicago, added after the design was fixed, has none.

*Alt text:* Left, a map of Boston's 14 neighborhoods shaded from light to dark blue, darkest in Dorchester and Roxbury, with dots at each neighborhood connected by lines to bordering neighborhoods and dashed lines to 2 water points. Right, horizontal bars for Boston, Columbus, and Chicago showing years of available data, blue blocks for unseen test seasons (3 for Boston, 1 for Columbus, 4 for Chicago), and gray blocks for the Boston and Columbus development seasons.

**Figure 2.** Forecast error relative to persistence by city, model, and horizon, unseen test seasons: A) Boston, Massachusetts, 2022–2025; B) Columbus, Ohio, 2024–2025; C) Chicago, Illinois, 2022–2026. Models shown are the graph neural network, the same model with the map removed, the tuned LSTM, tuned Cola-GNN, the seasonal blend (no learning), and the FluSight baseline. Relative weighted interval score (WIS) is each model's mean WIS divided by that of persistence (next week equals this week); values below the dotted line at 1 indicate less error than persistence. Learned models were tuned with an equal budget. LSTM, long short-term memory network; WIS, weighted interval score.

*Alt text:* Three dot plots, one per city, of relative error at 1, 2, and 4 weeks ahead for 6 models. In Boston and Columbus the graph neural network is lowest at every horizon,

between 0.81 and 0.87 at 1 and 2 weeks and near 0.72 to 0.75 at 4 weeks. In Chicago it is lowest at every horizon, near 0.81 to 0.83, and the FluSight baseline sits near 1.

**Figure 3.** Week-level differences in forecast error between the graph neural network and 3 comparison models, by city, season, and horizon, Boston, Massachusetts (2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (2022–2026). A) The same model with the neighborhood map removed. B) Tuned long short-term memory network (LSTM). C) Tuned Cola-GNN. Points show the difference in mean weighted interval score (WIS) (graph neural network minus comparison) as a percentage of the comparison model's mean WIS, so that Chicago's activity levels and the other cities' rates share one scale, and bars show 95% confidence intervals from a block bootstrap over weeks. Negative values favor the graph neural network. Filled points are differences with a Holm-adjusted 2-sided Diebold–Mariano *P* value below .05.

*Alt text:* Three forest plots with rows for Boston 2022–2023, 2023–2024, and 2024–2025, Columbus 2024–2025, and Chicago 2022–2023 to 2025–2026, and a point with an error bar for each of 3 horizons. In the first panel most Boston points lie left of zero, with filled points for all Boston seasons at 1 week and for 2023–2024 at 2 and 4 weeks; every Chicago point is left of zero, filled at 1 week in all 4 seasons. In the LSTM panel, 1-week points sit on zero for every Boston season, while Chicago points are left of zero and mostly filled. In the Cola-GNN panel, Boston points are left of zero but open.

**Figure 4.** Reduction in forecast error from adding the neighborhood map, by neighborhood and horizon, Boston, Massachusetts, June 2022–May 2025. A) 1 week ahead. B) 2 weeks ahead. C) 4 weeks ahead. Shading shows the percentage by which the graph neural network's mean weighted interval score (WIS) was lower than that of the same model with the map removed, over the 3 unseen seasons. Darker blue means a larger reduction; no neighborhood had higher error with the map at any horizon. WIS, weighted interval score.

*Alt text:* Three maps of Boston's neighborhoods in shades of blue. At 1 week the shading is darkest, with the largest reductions (12% to 15%) in West Roxbury, the South End, and

Roslindale; at 2 and 4 weeks the shading is lighter, mostly 2% to 8%. Every neighborhood is shaded, meaning the map reduced error everywhere.

**Figure 5.** Example forecasts and calibration, Boston, Massachusetts, 2022–2025. A) Percentage of observed values inside the 50% (open circles) and 95% (filled circles) forecast intervals for the graph neural network, tuned LSTM, tuned Cola-GNN, and persistence by horizon, 14 neighborhoods, June 2022–May 2025. Dotted lines mark the nominal 95% and 50% coverage. B–D) Forecasts for Dorchester, the neighborhood with the highest rates, June 2024–May 2025, made 1 (B), 2 (C), and 4 (D) weeks ahead, showing the graph neural network's median forecast, 50% and 95% forecast intervals, persistence, and observed emergency department visit rates for influenza-like illness per 100,000 residents. In Dorchester that season, relative WIS (the graph neural network's mean WIS divided by that of persistence) was 0.85, 0.69, and 0.51 at 1, 2, and 4 weeks, against 0.82, 0.81, and 0.72 for all 14 neighborhoods over 3 seasons (Table 2), so panel D is a better-than-typical case. ED, emergency department; ILI, influenza-like illness; LSTM, long short-term memory network; WIS, weighted interval score.

*Alt text:* Left, coverage dots for 4 models cluster at 91% to 95% for the 95% intervals and 50% to 55% for the 50% intervals. Right, 3 stacked weekly time series for Dorchester from mid-2024 to mid-2025, with observed dots rising from about 13 to nearly 300 per 100,000 in early February 2025 and falling by April. At 1 week the model's median line and the persistence line nearly overlap, and the median peaks near 250. At 2 weeks the median peaks near 225 and persistence trails 2 weeks behind. At 4 weeks the median peaks near 165, well below the observed peak, while persistence trails a month behind and stays near 290 into March, after the observed rates have fallen.

**Figure 6.** Severity band of each neighborhood-week, observed and forecast 2 weeks ahead, 14 neighborhoods, Boston, Massachusetts, June 2022–May 2025 (3 unseen seasons; 2,103 neighborhood-weeks). A) Graph neural network. B) Persistence (next week equals this week). Each forecast is placed in a band by its best guess (the median). Bands use each

neighborhood's Moving Epidemic Method intensity thresholds, fitted only on seasons before each test season: low is below the 50th-percentile threshold, moderate from the 50th to below the 90th, high from the 90th to below the 98th, and very high at or above the 98th. Each cell gives the number of neighborhood-weeks with that observed band (row) and forecast band (column); shading shows the share of each observed band's weeks, so the diagonal shows how often the band was forecast correctly.

*Alt text:* Two 4-by-4 grids of counts, rows for the observed band and columns for the forecast band, from low to very high. In the graph neural network's grid, most low weeks are forecast low, but most high and very high weeks are forecast moderate or low and no week is forecast very high. In the persistence grid, counts are spread more evenly across the bands for the high and very high rows, with more of those weeks forecast high or very high and more low weeks forecast above low.

# Figures

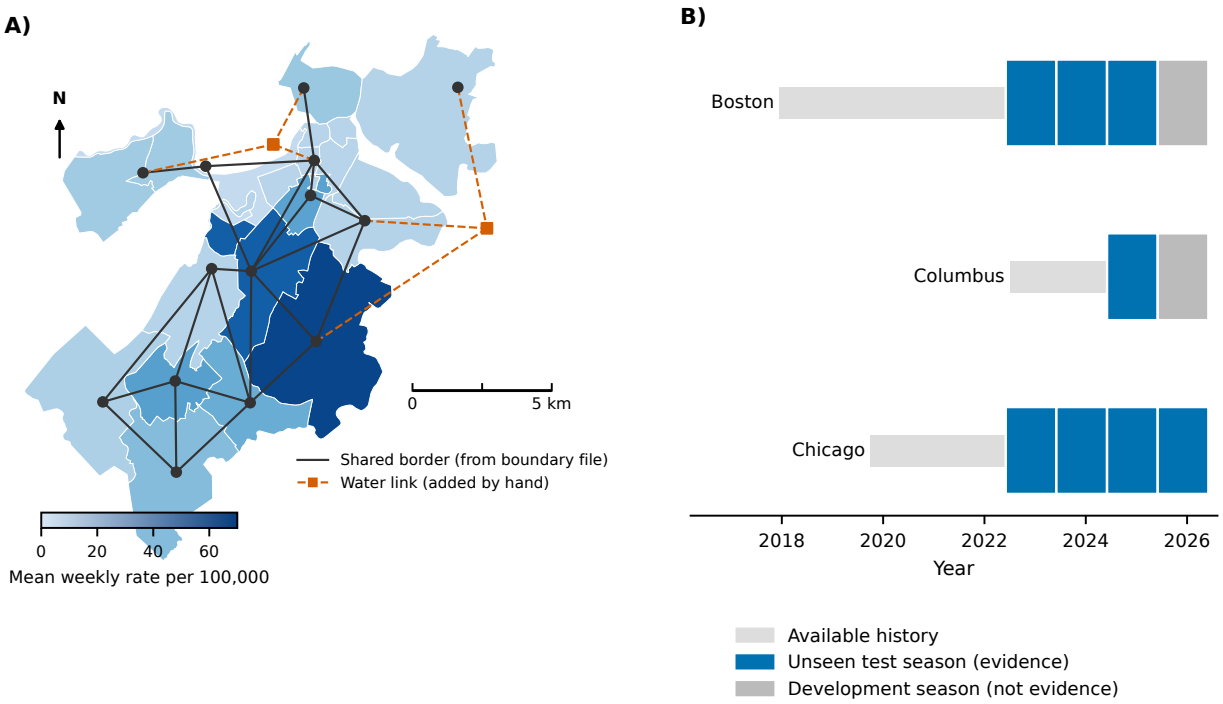



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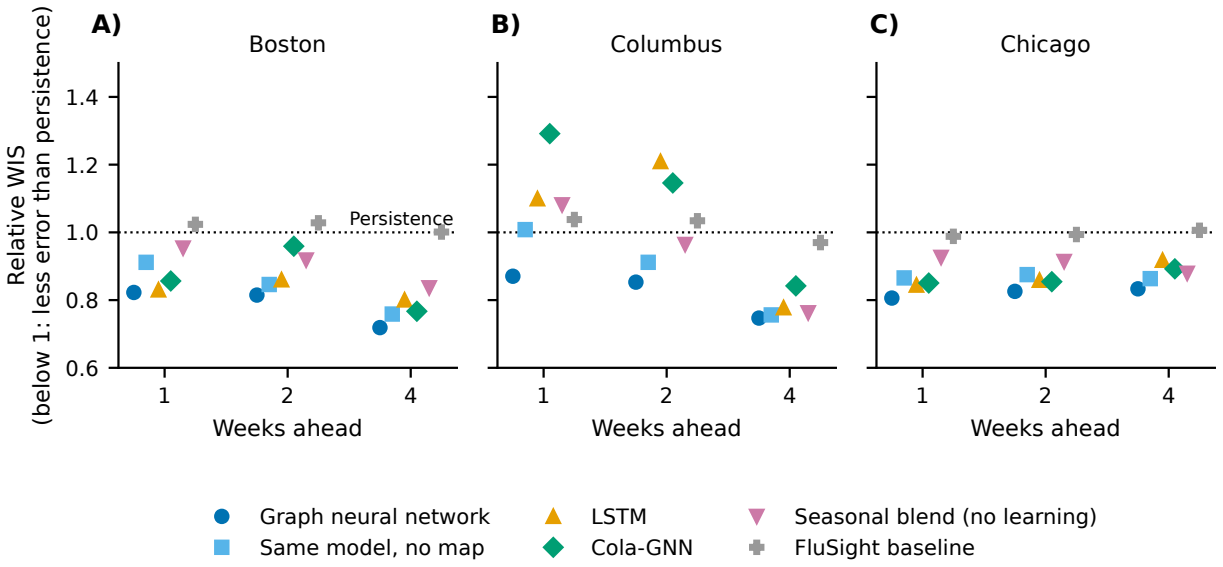



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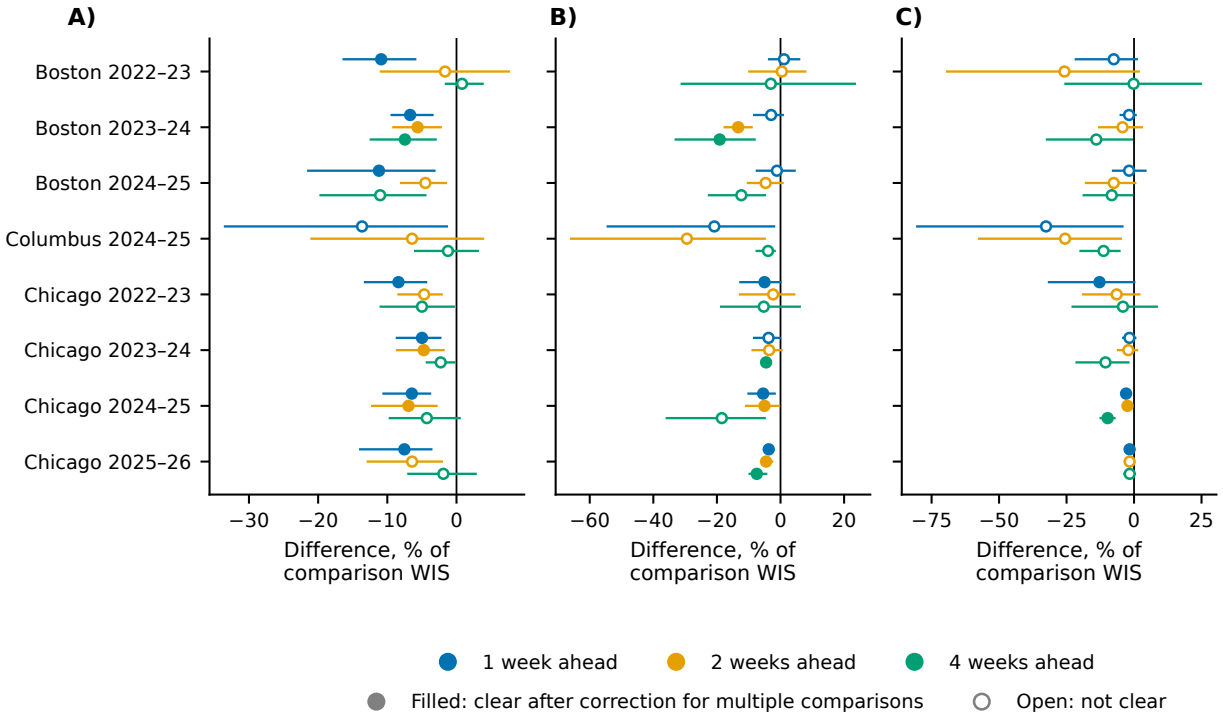



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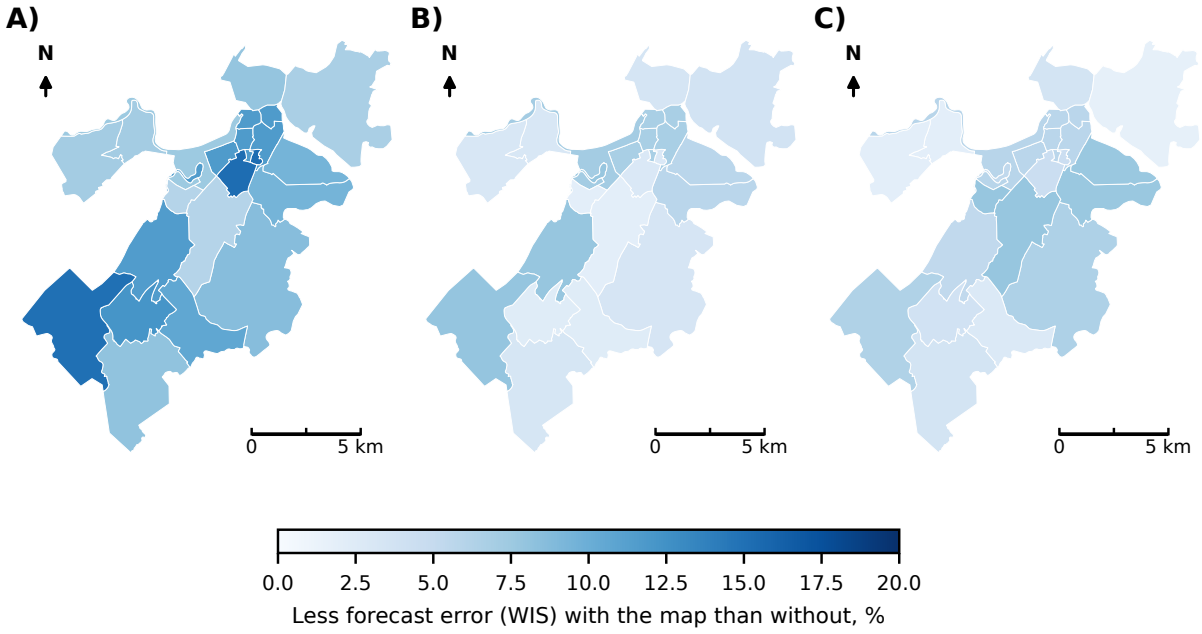



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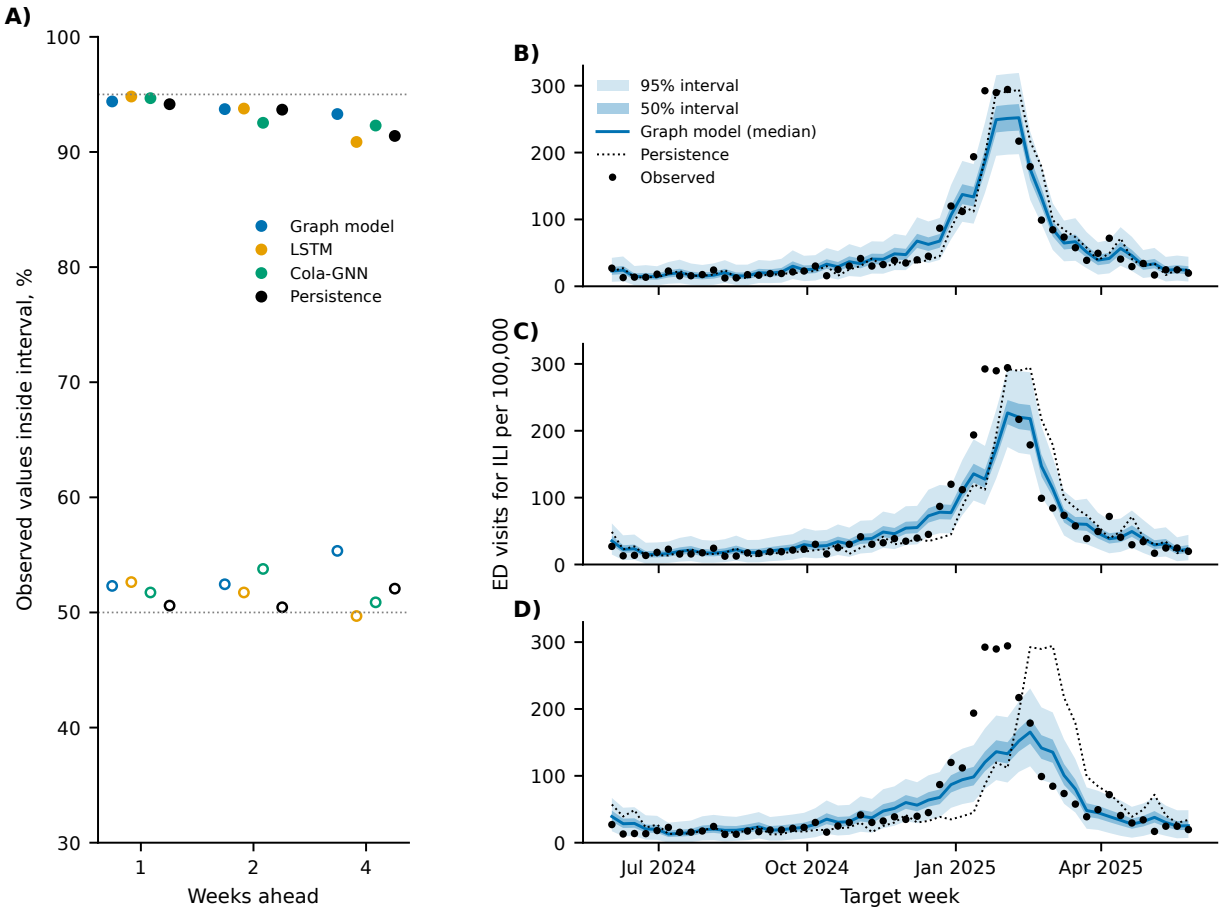



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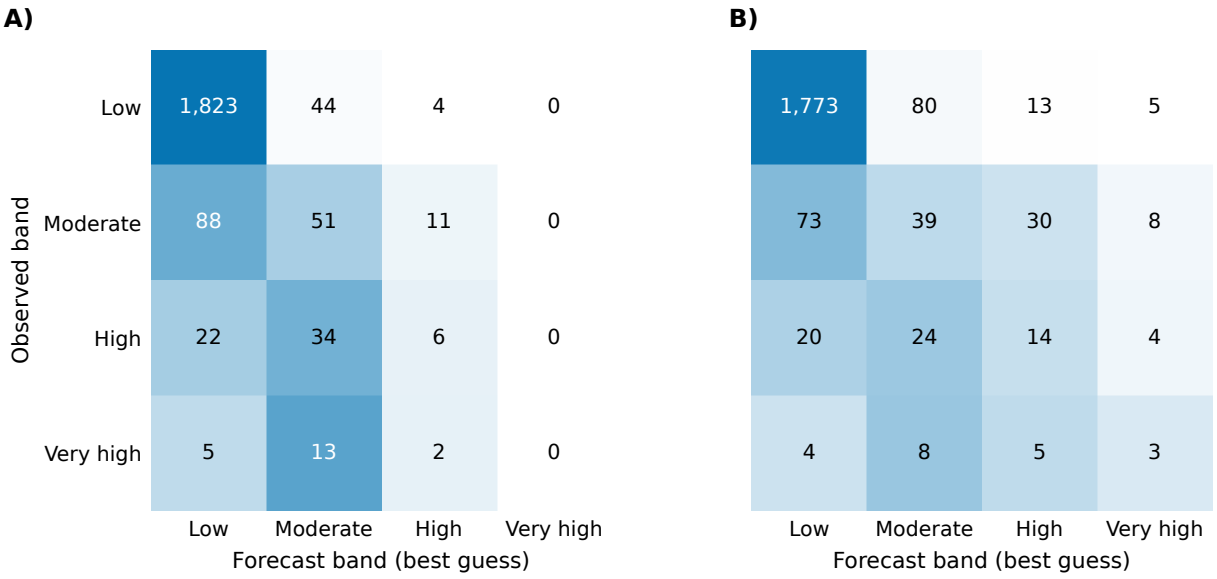



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